313339

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions :

(1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Assume suitable data, if necessary.

(6) Use of Non-programmable Electronic Pocket Calculator is permissible.

(7) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

(8) Use of steam tables, logarithmic, Mollier's chart is permitted.

Marks

1. Attempt any FIVE of the following : 10

(a) Name the salts which cause the temporary and permanent hardness in water.

(b) Define ton of refrigeration.

(c) Define wet bulb and dry bulb temperature.

(d) What is R-22 ? Give its properties.

(e) Name four important refrigerants used in industries.

(f) Define hard water and soft water.

(g) Give any two advantages of thermic fluid over steam.

2. Attempt any THREE of the following : 12

(a) Explain the working of forced draft cooling tower with labelled diagram.

(b) State selection criteria for refrigerant.

(c) A refrigeration system is operated in between 40 °C and –15 °C, the capacity of machine is 10 tonnes. Find out C.O.P.

(d) State duties and responsibilities of plant maintenance department.

3. Attempt any THREE of the following : 12

(a) Explain Zeolite process of water treatment with neat diagram.

(b) Explain vapour compression refrigeration cycle.

(c) State any eight industrial uses of air.

(d) Explain humidification and dehumidification.

4. Attempt any THREE of the following : 12

(a) Give uses of the following :

(i) Steam trap

(ii) Water level indicator

(iii) Safety valve

(iv) Eonomizer

(b) State reactions involved in Ion exchange process.

(c) Explain using a flow sheet the process of getting compressed air.

(d) Explain shutdown maintenance in detail.

(e) Explain Reverse Carnot Cycle in detail.

5. Attempt any TWO of the following : 12

(a) Discuss Indian Boiler Act regarding following points :

(i) Registration of Boiler

(ii) Certificate of Renewal

(iii) Transfer of Boiler

(iv) Penalty

(b) Calculate specific enthalpy and entropy of 1 kg of steam at pressure 10 bar.

(i) When steam is dry saturated

(ii) When steam is 75% dry

(c) State types of maintenance and explain preventive and predictive maintenance.

6. Attempt any TWO of the following : 12

(a) Explain hot lime soda method for water treatment with neat diagram.

(b) Define :

(i) Scale and Sludge formation

(ii) Caustic Embrittlement

(iii) Priming

(iv) Foaming

(v) Temporary hardness

(vi) Permanent Hardness

(c) Explain construction and working of Lancashire Boiler with neat diagram.

_____